INCH-POUND
MIL-D-24304B(SH)
AMENDMENT 2
22 February 1991
SUPERSEDING
AMENDMENT 1
24 January 1990

MILITARY SPECIFICATION

DIFFERENTIAL PRESSURE TRANSDUCER EQUIPMENT
(ELECTRICAL) (NAVAL SHIPBOARD USE)

This amendment forms a part of MIL-D-24304B(SH), dated 5 July 1988, and is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* 1.2.1: Delete designations "D" and "SG" in their entirety.

PAGE 2

* 1.2.1.2: Delete in its entirety.

* 1.2.1.3: Delete in its entirety.

* 1.2.1.4: Delete and substitute:

"1.2.1.4 Application. The fluid pressure to be measured shall be indicated as follows:

GP - Steam, oil, fresh water, condensate, and other gases
SW - Seawater
FG - Flue gas and ammonia
OX - Oxygen

PAGE 3

* 2.1.1, under SPECIFICATIONS, MILITARY, Delete: "MIL-C-52211 - Components and Assemblies for Industrial Gas Production, Storage and Transport Equipment, Packaging of."

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2.1.2: Delete and substitute:

"2.1.2 Other Government drawings and publications. The following other Governmnet drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

NAVAL SHIPS SYSTEMS COMMAND (NAVSHIPS)
810-1385846 - Oxygen System, Submarine Replenishment.

NAVAL SEA SYSTEMS COMMAND (NAVSEA)
803-5959187 - Pressure/Diefferential Pressure Transducer Equipment Application/ Selection/Installation Guide.

PUBLICATION

NAVSEA
NAVSHIPS 0900-LP-001-7000 - Fabrication and Inspection of Brazed Piping Systems.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)"

PAGE 6

* TABLE I: Delete and substitute:

"TABLE I. Material.

Sensing element and wetted parts	Steam, oil, fresh water, condensate, and other gases	Seawater	Service fluid flue gas and ammonia	Oxygen
CRES 304L 316L and 321	X	---	X	---

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TABLE I. Material. - Continued

Sensing element and wetted parts	Steam, oil, fresh water condensate, and other gases	Seawater	Service fluid flue gas and ammonia	Oxygen
CRES 17-4 PH 17-7 PH 15-5 PH	X	---	---	---
Monel and K-Monel	X	X	---	X
Inconel 600 and X750	X	---	---	---
Inconel 625 and 718	X	X	---	---
Hastelloy C276	X	X	X	---
Titanium CP and 6Al-4V	X	X	---	---
CuNi 70/30	X	X	---	X
Ni span	X	---	---	---
Tantalum	X	X	---	---

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* TABLE II: Delete and substitute:

"TABLE II. Ranges.

Type PDT	
Differential pressure ranges (PSID)	Differential pressure water column ranges (inches WCD)
0-15 0-30 0-60 0-100 0-200 0-400 0-600	0-10 0-30 0-60 0-150 0-300

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PAGE 8

* 3.5.5 (f), line 2: Delete "2.0" and substitute "1.5".

PAGE 9

3.5.6, after last sentence, Add: "The electrical receptacle may be designed to include built-in radio frequency suppression elements."

* 3.5.9, lines 4 and 5: Delete "Oil may be present only in detectors for SW, OS, and FG applications."

PAGE 10

* 3.5.9 (a), line 2: Delete "MIL-C-52211" and substitute "MIL-E-17555 (see 5.1)."

* 3.5.9 (b), line 1: Delete "GW" and substitute "GP, SW, and FG".

* 3.5.6 (c): Delete in its entirety.

* 3.5.9 (d): Delete in its entirety.

* 3.5.9.1, line 1: Delete "GW" and substitute "GP".

3.5.10: Delete and substitute:

"3.5.10 Welding and brazing. Internal pressure containing parts shall be joined by welding or brazing. Welding shall be in accordance with MIL-STD-278 and brazing in accordance with NAVSHIPS 0900-LP-001-7000.

3.5.10.1 General applications. Joints shall be either welded or nicrobrazed. Depending on the pressure range, MIL-STD-278, class P-1 or P-2 shall apply for welding. Class P-3 shall apply for nicrobrazing.

3.5.10.2 Oxygen applications. Joints shall be welded for all pressure ranges. MIL-STD-278, class P-1 shall apply."

* 3.5.12, line 2: Delete "GW" and substitute "GP".

Add new paragraph 3.5.14:

"3.5.14 Label plate. A metallic label plate with engraved or stamped markings shall be permanently affixed to the outside of each differential pressure transducer. The label plate shall not

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be placed on the underside of the mounting surface or on any portion, such as an access cover, which may be separated from the body of the differential pressure transducer. As a minimum, the label plate shall contain the following information:

(a) Differential pressure transducer
(b) Manufacturers name
(c) Manufacturers part number
(d) Contract number
(e) Date of manufacture
(f) Serial number
(g) Government designation (see 1.2.1)
(h) Differential pressure range"

PAGE 13

* 3.7.23 (d) (1): Delete and substitute:

"(1) The emission signal shall be measured only on the a.c. power leads of the test sample."

Add as new paragraph 3.9:

"3.9 Technical data. (See 6.2.2 and appendix)."

PAGE 14

4.3, after last sentence, Add: "Except for electromagnetic interference emission and susceptibility, all examinations and tests shall be conducted utilizing a single transducer, one for each range (see 4.3.1), and in the order listed in table III. The electromagnetic interference emission and susceptibility test may be conducted on a separate transducer of the same design and differential pressure range. In addition, this test may be conducted concurrently with any of the other examinations and tests listed in table III. Deviations from the above procedure require the prior approval of NAVSEA."

PAGE 16

* 4.4, lines 4 and 5: Delete "be cause for rejection." and substitute "result in rejection of that transducer."

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* 4.5, line 4: Delete "no transducers shall" and substitute "that lot of transducers shall not".

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* 4.5, line 10: Delete "results of all repeated tests" and substitute "based on test results".

PAGE 18

4.8.3, Line 5: Delete "0.25" and substitute "1.00",
Line 10: Delete "0.25" and substitute "1.00".

PAGE 22

4.8.17: Delete and substitute:

"4.8.17 Differential pressure whip. Before performing the differential pressure whip test, a reference measurement shall be made (see 4.8.2). The test shall be conducted on a system capable of applying a periodic differential pressure change of not more than 20 percent to not less than 80 percent of transducer span for a total of 260,000 cycles. The rate of cycling shall be 1.0 Hz + 0.5 Hz. Additionally, the time for that portion of the cycle with differential pressures greater than the average of the maximum and minimum excursion points shall be between 33 percent and 67 percent of the time for one complete cycle. The transducer shall be energized throughout the test. After completion of the differential pressure whip test, a reference measurement shall be made for comparison (see 4.8.2). Performance shall conform to 3.7.16."

4.8.18, Line 2: Delete "500" and substitute "50".

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6.2.2: Delete and substitute:

"6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
3.9	Drawings engineering	DI-DRPR-80651	
4.7.2	Special inspection equipment calibration procedures	DI-CMAN-80787	---

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(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5010.12-L., AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)"

LAST PAGE

At the end of document: Add attached Appendix, Engineering Drawings Technical Content Requirements.

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APPENDIX

Engineering Drawings Technical Content Requirements

10. SCOPE

10.1 Scope. This appendix contains the format and content preparation instructions for the development of the revision to a conformance verification drawing. It is not intended that each requirement contained herein should be applied to every type of instrumentation. Portions of this appendix are subject to deletion tailoring depending upon the material, construction and principle of operating requirements that are specified in the individual instrumentation specification or acquisition document. This appendix is mandatory only when data item description DI-DRPR-80651 is cited on the DD Form 1423.

20. APPLICABLE DOCUMENTS

20.1 Government documents. The following documents form a part of this appendix to the extent specified herein.

SPECIFICATION

MILITARY
MIL-I-45208 - Inspection System Requirements.

STANDARD

MILITARY
MIL-STD-17-1 - Mechanical Symbols (Other Than Aeronautical, Aerospacecraft and Spacecraft Use) Part-1.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

20.2 Other publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the document which is DoD adopted is that listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

Y14.1 - Drawing Sheet Size and Format.
IEEE 200 - Standard Reference Designations for Electrical and Electronics Parts and Equipment.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

30. CONFORMANCE VERIFICATION DRAWING FOR INSTRUMENTATION

30.1 Purpose. The conformance verification drawing contains the information necessary to verify that the instrumentation meets the requirements specified in the applicable instrumentation specification and acquisition document.

30.2 Content and format. One conformance verification drawing shall be developed for each specific type of instrumentation. It shall include all ranges, sizes, connections and other variations. The conformance verification drawing shall include the following minimum information (except as specified in 10.1) and shall be developed to the following format:

30.3 Descriptive data.

(a) Instrument identification numbering system for instrumentation. This numbering system shall include, but not be restricted to, the classification variables.
(b) Instrumentation identification number system for replaceable parts.
(c) Size, operating data, ranges, scale markings, and other data for proper selection.
(d) Test approval data, presenting the following information in tabular form.
    (1) Specification classification (unique alphanumeric variables to designate instruments).
    (2) Test report number and date.

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(3) Facility where test was conducted.
(4) Authorized Government activity (NAVSEA) approval letter and date.

(e) A statement that instrumentation is in accordance with the requirements of the applicable instrumentation specification or acquisition document and to referenced specifications.

(f) Conformance verification drawing acceptance data, presenting the following information in tabular form:
(1) Authorized Government activity (NAVSEA) acceptance letter and date.
(2) Revision number.

30.4 Details of construction.

(a) Two or more representative assembly views, as required, to show clearly the details of the design, construction, and assembly of the instrumentation and to identify each part and its location. Identification of parts shall correspond to the list of materials. Assembly shall show how all mechanical parts are joined or attached.
(b) Sectional views or notes as necessary to show internal details.
(c) Details such as entrance provisions, gaskets, fastening techniques, welding symbols, mounting requirements, and other details as applicable.
(d) Module enclosure requirements for separately mounted modules.
(e) Dimensions required to ensure interchangeability.
(f) Schematic of indicating system.
(g) Description of the nature and purpose of any adjustments.
(h) Welding procedures with acceptance data, including acceptance letter, report numbers and dates.
(i) Any special features.
(j) Complete weight.
(k) Location, size, and type of connections.
(l) Identification of quality control documents which show conformance with MIL-I-45208 or the quality control specification that is listed in the applicable instrumentation specification or acquisition document. The acceptance letter shall be referenced.

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30.5 Parts list. The following information shall be presented in tabular form:

(a) Item number (corresponding to flag number identifying the part on the conformance verification drawing).
(b) Quantity of each part required by assembly.
(c) Name of part with sufficient information to readily identify the part (for example, screws: thread size, length and type of head, shall be specified).
(d) Material of part.
(e) Material specification (military, federal, or commercial specification number or Government activity (NAVSEA) drawing number).

NOTE: When substitution of a material specification is made, it is the responsibility of the contractor to provide written documentation to substantiate that the substituted material is equivalent to the specified material.

(f) Type, class, grade, size, military designation, or other classification of any referenced specification.
(g) Part number or identification assigned by assembly supplier.
(h) Name of actual manufacturer of part (when applicable).
(i) Part number or identification assigned by part supplier (when applicable).
(j) Onboard repair parts. Parts that are appropriate for, or are supplied as, onboard repair parts shall be indicated by a symbol in this column.
(k) Remarks column. Finishes, platings, or coatings along with the applicable specification or other requirements should be specified in this column.

30.6 Table for special tools. Special tools required for the instrumentation shall be presented in the following tabular form:

(a) Item number (corresponding to flag number identifying the tool on the conformance verification drawing).
(b) Quantity of each tool required by instrument.
(c) Name (description) or tool (include generic name).
(d) Tool specification (military, federal, or commercial specification number or Government activity (NAVSEA) drawing number).

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(e) Tool number or identification assigned by assembly supplier.
(f) Name of actual manufacturer of tool (when applicable).
(g) Tool number or identification assigned by tool supplier (when applicable).
(h) Description of tools application.
(i) Remarks column. Special techniques or other usage requirements should be explained in this column.

30.7 Fluid and electrical system schematics. Fluid (piping) and electrical schematics shall be included in the verification conformance drawing when applicable. A complete schematic shall contain all the parts in the fluid or electrical systems. When confusion over its function could result, a simplified schematic containing only the major components shall also be provided. This simplified schematic may be presented in block diagram format.

(a) Complete fluid system schematic. A single schematic shall present clearly the operation and functions of the fluid system within the instrumentation. This schematic shall contain all parts (including valves, fittings, hoses, and tubing) which make up the piping or fluid system. The following features shall be incorporated into the schematic:
(1) A simplified, clear schematic in preference to one showing the physical placement of parts.
(2) A thin, broken line shall be used to represent the boundaries of each subassembly.
(3) Arrows adjacent to the parts shall indicate direction of flow.
(4) Each part shall be identified by the schematic part designation. The numbers shall be assigned in a logical sequence observing flow paths through the system.
(5) All inlet and outlet ports shall be identified.
(6) The methods of attachment for each part shall be designated by an appropriate label or symbol.
(7) The symbol used to designate a part shall be in accordance with MIL-STD-17-1 when applicable.
(8) A brief description of the fluid system operation shall be provided.
(9) Maximum operating pressure of the system and maximum pressure drop through the system (when operating at maximum pressure) shall be specified.
(10) Table for pressure rating and connection of fluid system parts. Supplementary information

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on parts found in the fluid (piping) system schematic shall be contained in tabular form One column shall contain the schematic part designation for each fluid system part; for example, valve V-1 and other similar designations. The other columns shall contain the following information:

a. Item number.
b. Description.
c. Type (size).
d. Pressure ratings (including operating, proof, and burst).
e. Method of attachment, including, as applicable, brazing or welding procedure, bonding agent, and seal (gasket, o-ring, and so forth).
f. End connections, fittings, and adapters.
g. Relief valve ratings (include cracking and set pressure).

(b) Complete electrical system schematic. A single schematic shall represent clearly the operation and the function of the electrical circuitry within the instrumentation. The schematic shall contain all parts (including components, connectors, alarms, and so forth) which make up the electrical system. The following features shall be incorporated into the schematic:

(1) In preparation of the schematic, emphasis shall be placed on simplicity and ease of understanding of circuit operation. Physical placement of components and connecting wiring may be ignored in the interest of simplicity and clarity of this diagram.

(2) A thin, broken line shall be used to represent the boundaries of each unit or subassembly. Terminals, to which external connections are made, shall be shown within these boundaries, with the numbers, markings, type of signal, power and ground, as appropriate.

(3) Each part (such as resistors, capacitors, relays, and so forth) shall be given a unique reference designation consisting of a letter denoting the type of part (as required by ANSI/IEEE 200) and a number assigned consecutively. The numbers shall be assigned in a logical sequence of electrical current or signal flow through the circuit.

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(4) In addition to where the reference designation parts not conforming to a military specification, where permitted, shall have the following information noted adjacent to the part. Where numerical values are given, a code shall be noted to designate the units used:
   a. Resistors - Resistance, power rating, and tolerances. If variable, an arrow to indicate clockwise rotation of the control shaft.
   b. Capacitors - Capacitance, voltage rating and tolerance.
   c. Reactors - Inductance and voltage rating.
   d. Semiconductors - Type number (JEDDC number is adequate).
   e. Integrated circuits - Operational symbol diagram of input-output relationship, terminal numbering corresponding to a representative schematic and type number.

(5) Supply voltages, phases, and frequencies and transformer terminal voltage shall be indicated and labeled as to purpose.

(6) Table for troubleshooting of electrical system. The content table shall include each test point, as identified on the electrical system schematic, with voltage, waveform or other electrical parameter that should be measured at each test point.

30.8 Selection and installation considerations.

(a) Performance data.
   (1) Accuracy (in percent of span).
   (2) Shock and vibration classification.
   (3) Degree of water tightness of the enclosure.
   (4) Electromagnetic interference and pulse susceptibility.

(b) Dimensional outline of the instrumentation showing overall and principle dimensions in sufficient detail to establish space requirements in all directions necessary for installation, servicing, exclusive of space required for operator observation of the indication.

(c) Special considerations which may affect selection or installation.
   (1) Ambient temperature range.
   (2) Calibration points and adjustments.

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(3) Orientation.
(4) Location of instrumentation relative to vibrating equipment.
(5) Protection of the instrumentation from pulsations and spikes in the parameter being measured.
(6) Selection of the instrumentation range relative to the operating range of the system.
(7) Application for each type connection.
(8) Cleaning procedure or reference to the cleaning procedure used.
(9) Selection of the instrumentation for compatibility (materials, temperature, pressure, and so forth) with the ambient environment and with the parameter being measured.

30.9 Drawing format.

(a) Unless otherwise approved by the authorized Government activity, a maximum of three sheets shall be allotted for single functioned system instrumentation and a maximum of ten sheets for instrumentation containing a multiple functioned system. A single functioned system is one that performs only one operation such as: expands/contracts a pressure elastic element, steps up/steps down the voltage, conditions one electrical signal, or winds/unwinds a bimetallic element. A multiple functioned system is one that contains two or more single functioned systems.
(b) Each sheet shall be zoned.
(c) Title block shall be included on each drawing sheet and shall include the following information:
(1) Title, drawing number, and revision letter. Each sheet contain the same title, drawing number and revision letter.
a. Title - The title shall consist of the name by which the instrumentation is known.
b. Drawing number - The drawing number shall consist of alphanumeric characters which may be separated by dashes or slashes. The total number of characters in the drawing number (including dashes/slashes) shall not exceed 15. Blank spaces are not permitted within the drawing number.
c. Revision letter - The revision letter shall denote the latest approved version of the

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drawing. The revision letter of conformance verification drawing shall not be changed until all the changes under that revision have been accepted in writing by the authorized Government activity (NAVSEA). No changes made to the conformance verification drawing shall be considered a revision until after the initial version of the conformance verification drawing has been accepted in writing by the authorized Government activity. After initial submittal of the conformance verification drawing, no changes shall be made during the initial drawing review process unless the change is either requested by the authorized Government activity or the change is documented in written correspondence by the contractor.

(2) Sheet_____ of_____.
(3) Tolerance on dimensions for fractions, decimals, and angles. Units of the dimensions specified on the conformance verification drawing.
(4) Contractor acceptance block (appropriate signatures and dates).
(5) Federal supply code for manufacturer (FSCM).
(6) Scale.
(7) Reference drawings.
(8) Manufacturers name and address.
(9) Drawing size.

(d) Revision block. The revision block shall be included on each sheet of the conformance verification drawing and shall contain the following information in tabular form:
(1) Revision letter.
(2) Description of revision.
(3) Acceptance letter serial number and originator identification.
(4) Acceptance date.

(e) Form. Sheet size and format not specified herein shall be in accordance with ANSI Y14.1.

(f) Classification designations. No Government security classification designation such as confidential or secret shall appear on the conformance verification drawing unless a particular classification is specified by the Government.

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40. CONFORMANCE VERIFICATION DRAWING ACCEPTANCE.

40.1 Acceptance. Acceptance shall be granted by the authorized Government activity only after the conformance verification drawing is found to meet all the requirements specified in 20 through 40.1.

50. Acquisition document instructions. The acquisition document should contain provisions that address submission, review, extension, disapproval, default, acceptance, and waiver of conformance verification drawings in addition to the effects on the delivery schedule due to delays in conformance verification drawing acceptance. NAVSEA shall be designated as the activity that accepts or disapproves the conformance verification drawing.

Preparing activity:
Navy - SH
(Project 6685-N861)

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